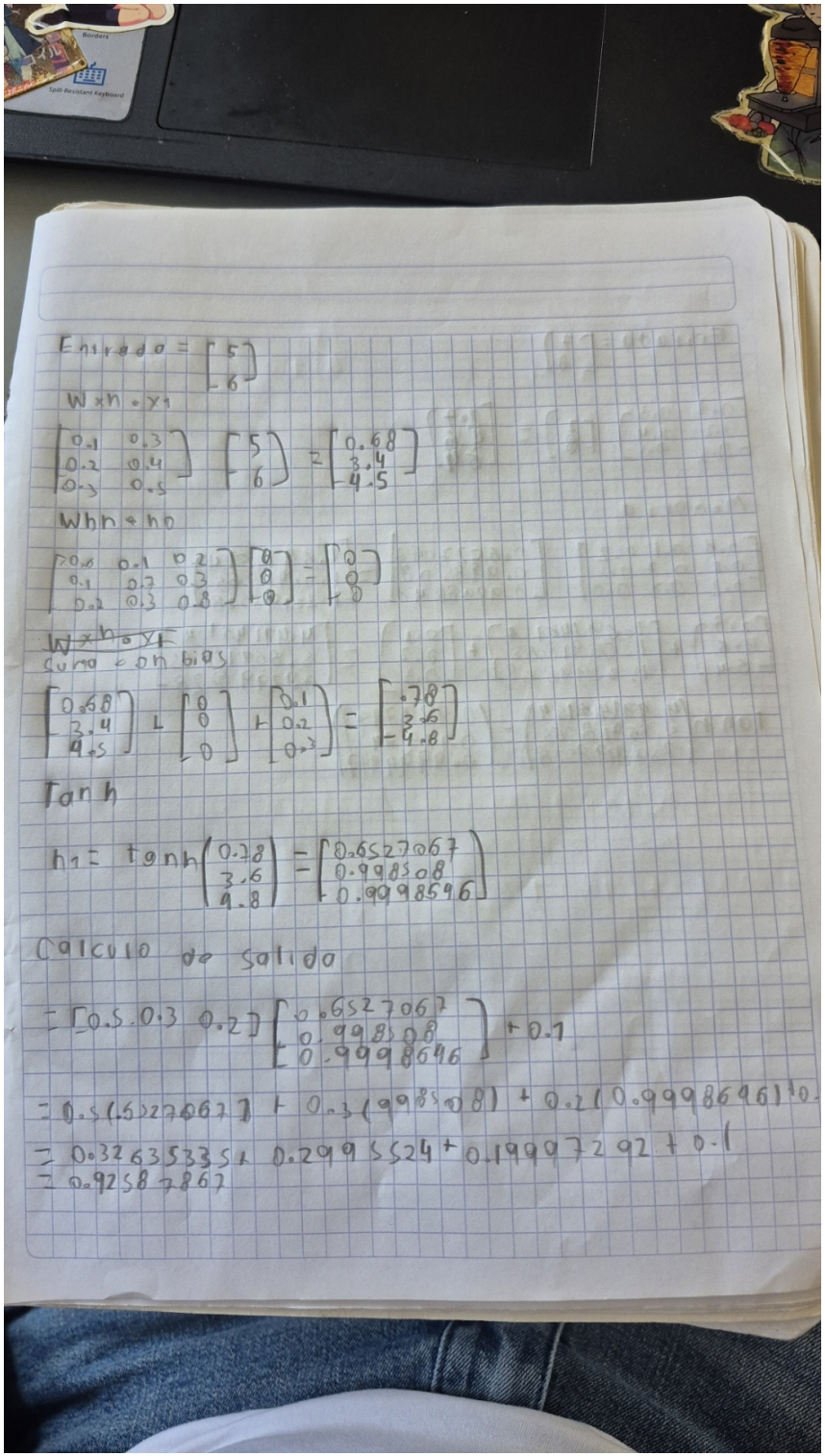


Entrada = $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

$W_{xh} \cdot x_1$

$\begin{bmatrix} 0.1 & 0.3 \\ 0.2 & 0.4 \\ 0.3 & 0.5 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.68 \\ 3.4 \\ 4.5 \end{bmatrix}$

$W_{hh} \cdot h_0$

$\begin{bmatrix} 0.8 & 0.1 & 0.2 \\ 0.1 & 0.7 & 0.3 \\ 0.2 & 0.3 & 0.8 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

~~$W_{xh} \cdot x_1$~~

Suma con bias

$\begin{bmatrix} 0.68 \\ 3.4 \\ 4.5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} = \begin{bmatrix} .78 \\ 3.6 \\ 4.8 \end{bmatrix}$

Tanh

$h_1 = \tanh\begin{pmatrix} 0.78 \\ 3.6 \\ 4.8 \end{pmatrix} = \begin{bmatrix} 0.65270067 \\ 0.9985008 \\ 0.99986496 \end{bmatrix}$

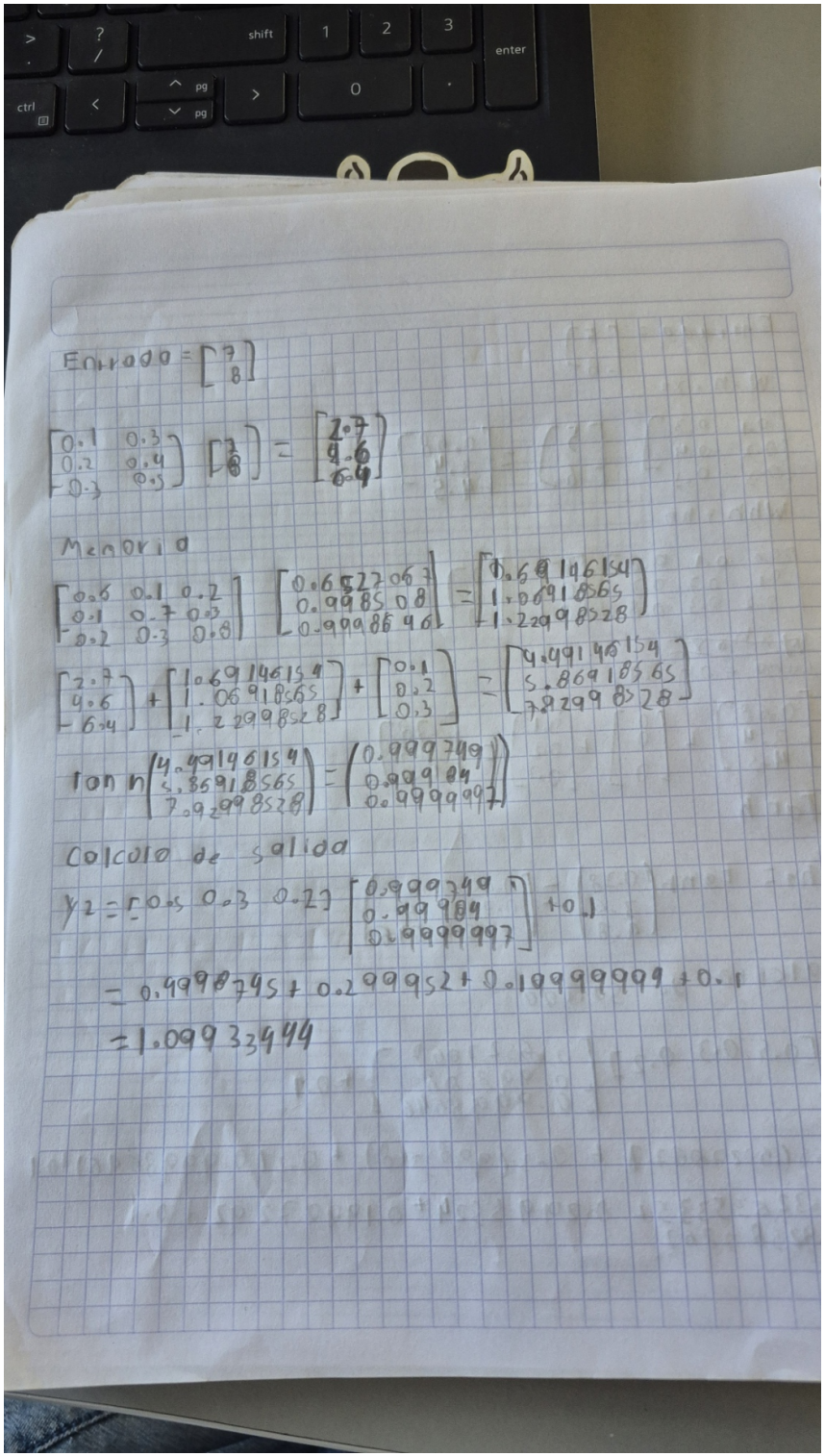
Calculo de salida

$= [0.5 \; 0.3 \; 0.2] \begin{bmatrix} 0.65270067 \\ 0.9985008 \\ 0.99986496 \end{bmatrix} + 0.1$

$= 0.5(0.65270067) + 0.3(0.9985008) + 0.2(0.99986496) + 0.1$

$= 0.32635335 + 0.2995524 + 0.19997292 + 0.1$

$= 0.92587867$


Entrada = [7 8]
[0.1 0.3; 0.2 0.4; 0.3 0.5] [7 8] = [2.7 4.6 6.4]
Memoria
[0.6 0.1 0.2; 0.1 0.7 0.3; 0.2 0.3 0.8] [0.6922067 0.9985088 0.9998646] = [0.6914654 1.0691856 1.2299828]
[2.7 4.6 6.4] + [1.0691465 1.0691856 1.2299828] + [0.1 0.2 0.3] = [4.4914654 5.8691856 7.8299828]
tanh([4.4914654 5.8691856 7.8299828]) = [0.9997749 0.9999784 0.9999997]
Calculo de salida
y2 = [0.5 0.3 0.2] [0.9997749 0.9999784 0.9999997] + 0.1
= 0.4998875 + 0.2999935 + 0.1999999 + 0.1
= 1.0998809